Smalltalk/CREXX

A Smalltalk-80 interpreter written in CREXX

Language Reference and User's Guide

Release 1 • Build 1
for CREXX 1.0.0 beta 3 and later

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This edition applies to Smalltalk/CREXX Release 1 (build 1, 10 October 2026), an interpreter for the Smalltalk-80 language written in CREXX Level B, and to all subsequent builds until otherwise indicated in new editions.

Smalltalk/CREXX is a hobbyist project. It is not a product of, and this manual is not a publication of, the authors or owners of any Smalltalk implementation. Smalltalk-80 and GNU Smalltalk are names of a language and a product, used here only to identify the language that this software implements and the implementation whose source syntax it follows.

CREXX is an open-source REXX implementation and is maintained by its own community. Smalltalk/CREXX needs it to run.

The project repository holds the interpreter source, the examples and the regression tests from which every statement in this manual can be checked. Comments on this publication may be addressed to the project.

Contents

Preface ... 1

Chapter 1. Introduction ... 2
- How Smalltalk/CREXX is built ... 2
- Status ... 2

Chapter 2. Installing and Running ... 4
- What you need ... 4
- Installation steps ... 4
- Running a program ... 4
- Starting the interpreter ... 4
- The regression tests ... 4

Chapter 3. The Smalltalk Language ... 5
- Comments and literals ... 5
- Variables ... 5
- Messages ... 5
- Cascades ... 6
- Assignment and statements ... 6
- Blocks ... 6
- Non-local return ... 6
- Control flow ... 7
- Top-level programs ... 7

Chapter 4. Classes and Objects ... 8
- Defining a class ... 8
- Reopening and extending classes ... 8
- Creating a class with messages ... 9
- Creating instances ... 9
- Inheritance and super ... 9
- Printing objects ... 10
- Metaclasses ... 10
- Class-side instance variables ... 11

Chapter 5. Numbers ... 12
- The number classes ... 12
- Arithmetic ... 12
- Large integers ... 12
- Floating-point numbers ... 13
- Reference ... 13

Chapter 6. Collections ... 14
- The collection classes ... 14
- Protocol common to all collections ... 14
- Arrays and OrderedCollections ... 15
- Dictionaries ... 15
- Sets and Bags ... 16
- Intervals ... 16

Chapter 7. Strings, Symbols and Characters ... 17
- Strings ... 17
- Symbols ... 17
- Characters ... 18

Chapter 8. Streams and the Transcript ... 19
- The Transcript ... 19
- Read streams ... 19
- Write streams ... 19
- Standard input ... 19

Chapter 9. Exceptions ... 20
- Handling ... 20
- Messages understood by an exception ... 20
- Signaling ... 20
- Cleaning up ... 21
- Your own exceptions ... 21

Chapter 10. Reflection and the System ... 22
- Asking about objects ... 22
- Asking about classes ... 22
- The Smalltalk dictionary ... 22

Chapter 11. Errors and Traces..... 24
Unhandled errors..... 24
Common messages..... 24
Parse errors..... 24
Chapter 12. How Smalltalk/CREXX Works..... 25
Objects..... 25
Classes and methods..... 25
The resolver..... 25
The evaluator..... 25
Primitives..... 25
Adding a primitive..... 25
The build..... 26
Chapter 13. Restrictions and Known Problems..... 27
Appendix A. Command Summary..... 28
Appendix B. Contents of the Repository..... 29
Appendix C. Class Hierarchy..... 30
Appendix D. Sample Programs..... 31

Preface

This manual introduces Smalltalk/CREXX, an interpreter for the Smalltalk-80 language written in CREXX. It tells you how to install and run the interpreter and describes the language it implements: the syntax of expressions and classes, the object model, the numeric tower, the collection classes, streams, exceptions and reflection.

It is written for the programmer who knows Smalltalk or another object-oriented language, and for the newcomer who wants to see what a pure object language looks like. Where Smalltalk/CREXX behaves as the classic systems do, this manual says little; where it differs, it says how.

How this manual is organized

- **Chapter 1, Introduction**, says what Smalltalk/CREXX is, how it is built and what it supports.
- **Chapter 2, Installing and Running**, covers the launcher, how programs are read and the regression tests.
- **Chapter 3, The Smalltalk Language**, describes literals, variables, the three kinds of message, cascades, blocks and control flow.
- **Chapter 4, Classes and Objects**, describes how classes are defined, class-side variables and methods, inheritance, super and metaclasses.
- **Chapter 5, Numbers**, covers integers of any size, fractions and floating-point numbers.
- **Chapter 6, Collections**, is the reference for Array, OrderedCollection, SortedCollection, Interval, Set, Bag and Dictionary.
- **Chapter 7, Strings, Symbols and Characters**, covers the text classes.
- **Chapter 8, Streams and the Transcript**, describes ReadStream, WriteStream and the Transcript.
- **Chapter 9, Exceptions**, describes on:do:, ensure:, signal, retry, resume and the exception classes.
- **Chapter 10, Reflection and the System**, covers the Smalltalk dictionary, perform: and the class protocol.
- **Chapter 11, Errors and Traces**, shows what the interpreter prints for an unhandled error.
- **Chapter 12, How Smalltalk/CREXX Works**, describes the object table, the resolver, the evaluator and the primitives for those who wish to extend the interpreter.
- **Chapter 13, Restrictions and Known Problems**, lists what does not work yet.
- **Appendix A** summarizes the command line, **Appendix B** lists the files of the repository, **Appendix C** shows the class hierarchy, **Appendix D** lists the sample programs.

Conventions

Program text is shown in monospaced type. In syntax descriptions, words in italics stand for values you supply and items in square brackets [ ] are optional. Smalltalk names are case sensitive: Total and total are different variables. By convention class names begin with a capital letter and variable and message names with a lower-case letter.

Examples of programs are shown in framed figures. In a figure that shows a run, the output of the interpreter follows the program text.

Chapter 1. Introduction

Smalltalk/CREXX is a Smalltalk-80 interpreter written in CREXX, the open-source REXX implementation. It runs programs written in the bracket syntax of GNU Smalltalk: classes are declared with Object subclass: Foo [...], methods are written inside the class body, and statements at the top level are run one after the other. The interpreter is a single CREXX source file, smalltalk.crexx, written in CREXX Level B, the statically typed level of the language, and a class library written in Smalltalk itself.

Smalltalk/CREXX gives its users:

- **A pure object model.** Everything is an object: integers, characters, classes and blocks. All computation is by sending messages, and the only control structures are messages to booleans, blocks and numbers.
- **Classes and metaclasses.** Every class is an instance of its metaclass; class-side methods, class variables and class-side instance variables are all supported.
- **Blocks.** Blocks are closures. They may be stored, passed, and returned from methods, and a ^ inside a block returns from the method that defined it, even through several levels of calls.
- **Numbers without limits.** Integers grow to any size, division of integers gives exact fractions, and floating-point numbers are available for the rest.
- **A class library.** Collections, strings, symbols, streams and exceptions are provided by library classes written in Smalltalk, loaded when the interpreter starts.
- **Exceptions.** Handlers with return, retry, resume, pass and outer; ensure: and ifCurtailed: blocks that run as the stack unwinds; user-defined exception classes.

How Smalltalk/CREXX is built

A source file is read once and parsed statement by statement into a tree held in parallel arrays. A resolver then walks the tree: it turns variable names into frame slots, instance-variable positions or global cells, and compiles the standard control-flow messages (ifTrue:, whileTrue:, to:do:, and:, or:, timesRepeat:, ifNil: and their relatives) into direct nodes when their arguments are literal blocks. The evaluator walks the resolved tree and sends messages through a method cache.

Objects are entries in a table of parallel arrays. SmallIntegers are not objects in the table: they are encoded in the reference itself. Common operations on numbers, strings, arrays and dictionaries are implemented as primitives in CREXX; the class library calls them through the pragma <primitive: name> and falls back to Smalltalk code when a primitive fails. Chapter 12 describes the design.

The interpreter is compiled by the CREXX compiler to a bytecode file, which the CREXX virtual machine runs. The first run of the launcher performs this compilation; later runs start in about a second, of which most is the loading of the class library.

Status

The figure below shows the major areas of the language and their status in this release.

Area	Content	Status
Syntax	Expressions, cascades, blocks, literals, brace arrays, class-body syntax, Eval, message-style class creation	complete
Object model	Classes, metaclasses, instance and class variables, class-side instance variables, super, extend	complete
Control flow	Inlined conditionals and loops, non-local return	complete
Numbers	SmallInteger, large integers, Fraction, Float, mixed arithmetic	complete
Collections	Array, OrderedCollection, SortedCollection, Interval, Set, Bag, Dictionary, Association	complete
Text	String, Symbol, Character	complete
Streams	ReadStream, WriteStream, ReadWriteStream, Transcript	complete
Exceptions	on:do:, ensure:, ifCurtailed:, signal, retry, return:, resume:, pass, outer, exception sets	complete

Area	Content	Status
Reflection	class, respondsTo:, perform:, selectors, instance variable names	most
Processes	Process, Semaphore, fork	not planned
Garbage collection	Objects are never freed	not planned
Bang (!) file format	The Smalltalk-80 chunk format	not planned

Figure 1. Smalltalk/CREXX language areas

Chapter 2. Installing and Running

What you need

Smalltalk/CREXX needs only the CREXX toolchain: the crexx driver together with the compiler (rxc), assembler (rxas) and virtual machine (rxvm) that it calls. CREXX 1.0.0 beta 3 was used to build and test the interpreter. The driver can be installed from a CREXX release package, or built from source with CMake.

The launcher script smalltalk is a POSIX shell script. The build script build.py needs Python 3, but only if you change the sources.

Installation steps

1. Install CREXX and make sure the crexx command is on your PATH. If it is not, set the environment variable CREXX to the full path of the driver.
2. Get the Smalltalk/CREXX repository (git clone https://github.com/mikebeer/smalltalk).
3. Run any program with the launcher: ./smalltalk examples/hello.st. The first run compiles the interpreter to bytecode and takes about half a minute. The compiled file, smalltalk.rxbin, is kept next to smalltalk.crexx.
4. Run the regression tests with ./tests/run_tests.sh. All of them should report ok.

Running a program

The launcher takes the names of source files. They are read and run in order, in one image: classes defined in the first file are visible in the second. With no file name the program is read from standard input. Output goes to standard output.

```
Transcript showCr: 'Hello, World!'.
(3 + 4) printNl.
#(1 2 3) printNl.
(1/3) + (1/6)

Hello, World!
7
(1 2 3 )
```

Figure 2. A first program, and its output

A top-level statement ends with a period. The value of a statement is discarded: unlike an interactive workspace, the interpreter does not print it. Use printNl or displayNl to show a value.

Starting the interpreter

When it starts, the interpreter creates the kernel classes, then reads the library files lib/kernel.st, lib/numbers.st, lib/collections.st, lib/streams.st and lib/exceptions.st, and only then reads your program. The library directory is found by the launcher; with the driver you pass it as -L directory.

The regression tests

The directory tests holds a number of programs with the output they must produce. The script tests/run_tests.sh runs every one and compares the output, error trace included. The option --update rewrites the expected files after a deliberate change.

Chapter 3. The Smalltalk Language

Comments and literals

A comment is text enclosed in double quotes. Literals are the following.

Kind	Examples	Class
Integers	42 -7 16r1F 2r1010 1e3	SmallInteger or LargePositiveInteger
Large integers	12345678901234567890123456789O	LargePositiveInteger, LargeNegativeInteger
Floats	3.14 -0.5 2.5e3 1.0e-3	FloatD
Characters	$a $ $'	Character
Strings	'hello' 'it''s'	String
Symbols	#foo #at:put: #+ #'hello world'	Symbol
Literal arrays	#(1 2 #three $4 (5 6) foo true)	Array
Brace arrays	{1+1. 'two'. #three}	Array (evaluated)
Constants	nil true false	UndefinedObject, True, False

Figure 3. Literals

Inside a literal array, a bare word is a symbol, nil, true and false are the constants, and a parenthesized group is a nested array. A brace array is different: each element is an expression, evaluated when the brace array is evaluated.

Variables

A variable name is a letter followed by letters, digits and underscores. There are four kinds of variable.

- **Temporaries.** Declared between vertical bars, | a b |, at the head of a method, of a block, or at the top level of a program. The variables of a block are local to the block.
- **Instance variables.** Declared in a class body; each instance has its own copy.
- **Class variables.** Declared in a class body as Name := expression. They are shared by the class, its subclasses and its instances.
- **Globals.** Classes, and the variables created with Smalltalk at:put:. An assignment to a name that is not declared anywhere creates a global.

The pseudo-variables self, super, nil, true and false cannot be assigned to. self is the receiver of the current message; super is self, with method lookup starting in the superclass of the class that holds the method.

Messages

All computation is by sending messages. There are three forms, and they bind in this order: unary messages first, then binary, then keyword.

Kind	Form	Example
Unary	receiver selector	5 factorial
Binary	receiver op argument	3 + 4 a -> b x >= 0
Keyword	receiver key1: a key2: b	dict at: 1 put: 'one'

Figure 4. The three kinds of message

```
(2 + 3 * 4) printNl.
(2 + (3 * 4)) printNl.
(3 max: 4 + 5) printNl.
#(3 1 2) asSortedCollection asArray printNl.

20
14
9
(1 2 3 )
```

Figure 5. Precedence

Binary messages are evaluated from left to right with no precedence among operators, so 2 + 3 * 4 is 20. Parentheses override the order.

Cascades

A semicolon sends another message to the receiver of the previous message. The value of the cascade is the value of the last message.

```
| oc |
oc := OrderedCollection new.
oc add: 3; add: 1; add: 2; yourself.
oc printNl.

OrderedCollection (3 1 2 )
```

Figure 6. A cascade

Assignment and statements

An assignment is written name := expression. A statement ends with a period; the value of an assignment is the value assigned, so a := b := 0 works. A caret, ^ expression, returns from a method.

Blocks

A block is a sequence of statements in square brackets, with optional parameters: [:x :y | x + y]. A block is an object of class BlockClosure. It is evaluated with value, value:, value:value:, value:value:value:, value:value:value:value: or valueWithArguments:, according to the number of arguments; sending it the wrong number gives the error wrong argument count.

A block keeps the variables of the context in which it was created. The method in the next figure returns a block that adds a fixed amount.

```
Object subclass: Maker [
    adder: k [ ^[:x | x + k] ]
]
| add5 |
add5 := Maker new adder: 5.
(add5 value: 10) printNl.
(add5 value: 20) printNl.

15
25
```

Figure 7. A closure

Non-local return

A ^ inside a block returns from the method in which the block was written, not just from the block. The loop in the next figure ends as soon as the element is found.

```
Object subclass: Finder [
    find: x in: coll [
        coll do: [:e | e = x ifTrue: [^'found']].
        ^'missing'
    ]
]
(Finder new find: 3 in: #(1 2 3)) displayNl.
(Finder new find: 9 in: #(1 2 3)) displayNl.

found
missing
```

Figure 8. Return from inside a loop

If the method has already returned when such a block runs, the return cannot happen and the interpreter reports BlockContext>>cannotReturn: and abandons the statement.

Control flow

Smalltalk has no special syntax for conditions and loops. They are messages. When the arguments are literal blocks the resolver compiles them in line, so they are as fast as built-in statements; otherwise they are ordinary sends, defined in the library, so that the same messages work with blocks held in variables.

Message	Receiver	Meaning
ifTrue: ifFalse: ifTrue:ifFalse: ifFalse:ifTrue:	Boolean	Conditional; the value is the value of the chosen block, or nil
and: or: & | not xor:	Boolean	Logical operations; and: and or: do not evaluate the block if the result is known
whileTrue: whileFalse: whileTrue whileFalse	Block	Loop while the receiver block answers true (false)
repeat	Block	Loop forever; leave with ^ or an exception
to:do: to:by:do:	Number	Count from the receiver to the limit with a step
timesRepeat:	Integer	Evaluate the block that many times
ifNil: ifNotNil: ifNil:ifNotNil:	Any object	Test for nil; the ifNotNil: block may take the object as argument
do: collect: select: reject: detect: inject:into:	Collection	Iterate over the elements (Chapter 6)

Figure 9. Control-flow messages

```
1 to: 3 do: [:i | Transcript show: i printString; show: ' '].
Transcript cr.
3 to: 1 by: -1 do: [:i | Transcript show: i printString; show: ' '].
Transcript cr.
| n |
n := 1.
[n < 100] whileTrue: [n := n * 2].
n printNl.

1 2 3
3 2 1
128
```

Figure 10. Loops

Top-level programs

A program is a sequence of class definitions and statements. Variables declared at the top level with | a b | live until the end of the file. A statement that fails with an error prints a trace (Chapter 11) and the program continues with the next statement. The construct Eval [statements] groups statements; it is accepted for compatibility with GNU Smalltalk.

Chapter 4. Classes and Objects

Defining a class

A class is declared with the message subclass: in bracket form. The body of the brackets holds instance-variable declarations, class-variable declarations and methods.

```
Object subclass: Account [
    | balance |
    Count := 0.
    Account class >> new [ Count := Count + 1. ^super new init ]
    Account class >> count [ ^Count ]
    init [ balance := 0 ]
    balance [ ^balance ]
    deposit: amount [
        amount <= 0 ifTrue: [^self error: 'Invalid amount'].
        balance := balance + amount
    ]
]
| a |
a := Account new.
a deposit: 50; deposit: 25.
a balance printNl.
Account new.
Account count printNl.
a deposit: -1.

75
2
Object: an Account error: Invalid amount
Account(Object)>>error:
Account>>deposit:
UndefinedObject>>executeStatements
```

Figure 11. A class with instance and class variables

Inside the body:

- | a b | declares instance variables.
- Name := expression. declares a class variable and gives it an initial value, which is computed once when the class is defined.
- selector [statements] defines an instance method. The pattern is a unary selector, a binary operator with one argument, or keywords with an argument each.
- Foo class >> selector [statements] defines a method on the class side. Inside the body of Foo, the class name may be written in this form.
- Foo class [| a b |] declares class-side instance variables, which each class (and each subclass) holds separately.
- <comment: 'text'> is accepted and ignored.

A method returns the value of the expression after ^. A method that does not execute a ^ returns self. Temporary variables of a method are declared at its start: | t1 t2 |.

Reopening and extending classes

Foo extend [...] adds methods and variables to an existing class, including the kernel classes, and Foo class extend [...] does the same for the class side. A single method may be added with Foo >> selector [...] or Foo class >> selector [...].

```
Integer extend [
    double [ ^self * 2 ]
    isPerfect [
        | sum |
        sum := 0.
        1 to: self - 1 do: [:i | self \\ i = 0 ifTrue: [sum := sum + i]].
        ^sum = self
    ]
]
5 double printNl.
((1 to: 500) select: [:n | n isPerfect]) printNl.

10
(6 28 496 )
```

Figure 12. Extending a kernel class

Methods added to a class are seen at once by every instance, including those created earlier. Instance variables added to a class are seen by new instances; instances that already exist are not changed.

Creating a class with messages

The classic form also works. The class is created by sending subclass:instanceVariableNames:classVariableNames:package: (or category:) to the superclass, and methods are added with extend.

```
Object subclass: #Pair
    instanceVariableNames: 'first second'
    classVariableNames: ''
    package: 'demo'.
Pair extend [
    first [ ^first ]
    first: x [ first := x ]
]
(Pair new first: 5; yourself) first printNl.

5
```

Figure 13. Message form

Creating instances

new and new: create instances. In this system new does not send initialize; the convention is to override the class-side new, as in the Account example above. new: n creates an indexable object with n slots, as for Array new: 5.

Inheritance and super

Messages are looked up in the class of the receiver, then in its superclass, and so on up to Object. If no method is found the message doesNotUnderstand: is sent to the receiver. A message to super looks up the method starting in the superclass of the class in which the method was written.

```
Object subclass: Animal [
    | name |
    Animal class >> named: aName [ ^self new setName: aName ]
    setName: aName [ name := aName ]
    speak [ ^self subclassResponsibility ]
    describe [ ^name, ' says ', self speak ]
]
Animal subclass: Dog [ speak [ ^'Woof' ] ]
Animal subclass: Cat [
    speak [ ^'Meow' ]
    describe [ ^super describe, '!' ]
]
(Dog named: 'Rex') describe displayNl.
(Cat named: 'Tom') describe displayNl.

Rex says Woof
Tom says Meow!
```

Figure 14. Inheritance

Printing objects

printNl prints printString and displayNl prints displayString. The printString of an object is built by sending it printOn: with a stream, so a class changes how it prints by overriding printOn:. displayString differs from printString for strings, symbols and characters (no quotes) and for collections (the elements are shown with displayString).

```
Object subclass: Point3 [
    | x y z |
    x: ax y: ay z: az [ x := ax. y := ay. z := az ]
    printOn: aStream [
        aStream nextPutAll: 'Point3('; print: x; nextPutAll: ', ';
            print: y; nextPutAll: ', '; print: z; nextPut: $)
    ]
]
(Point3 new x: 1 y: 2 z: 3) printNl.
(Array with: (Point3 new x: 0 y: 0 z: 0)) printNl.

Point3(1, 2, 3)
(Point3(0, 0, 0) )
```

Figure 15. Overriding printOn:

Metaclasses

A class is an object, and so has a class: its metaclass. The metaclass of Foo is written Foo class. The superclass of Object class is Class, so every class understands the messages of Class, Behavior and Object. Class-side methods are instance methods of the metaclass.

```
3 class printNl.
3 class class printNl.
3 class class class printNl.
3 class class class class printNl.
Integer superclass printNl.
Object superclass printNl.

SmallInteger
SmallInteger class
Metaclass
Metaclass class
Number
nil
```

Figure 16. Classes are objects

Class-side instance variables

Foo class [| count |] declares a variable that belongs to the class object itself. It is not shared with subclasses: each subclass has its own copy. This differs from a class variable, which is one cell shared by the class and all its subclasses.

Chapter 5. Numbers

The number classes

Number has the subclasses Integer, Fraction and Float. Integer has SmallInteger (up to 18 decimal digits) and LargePositiveInteger and LargeNegativeInteger. The transition between them is automatic and cannot be seen except by asking for the class. Float is represented by the class FloatD, a 64-bit floating-point number.

```
3 class printNl.
1000000000000 class printNl.
(2 raisedTo: 100) class printNl.
(2 raisedTo: 100) negated class printNl.
(1/2) class printNl.
3.5 class printNl.

SmallInteger
SmallInteger
LargePositiveInteger
LargeNegativeInteger
Fraction
FloatD
```

Figure 17. The number classes

Arithmetic

+, -, * and / are the four operations. Dividing two integers gives an integer if the division is exact and a Fraction otherwise; fractions are always reduced and a fraction with denominator 1 becomes an integer. // is integer division rounding toward minus infinity and \\ is the matching remainder; quo: and rem: round toward zero.

```
(7 / 2) printNl.
(8 / 2) printNl.
(7 // 2) printNl.
(-7 // 2) printNl.
(-7 \\ 2) printNl.
(-7 quo: 2) printNl.
(-7 rem: 2) printNl.
((1/3) + (1/6)) printNl.
((1/2) + 0.25) printNl.

7/2
4
3
-4
1
-3
-1
1/2
0.75
```

Figure 18. Division

When two numbers of different classes are combined, the one of lower generality is converted: SmallInteger and LargeInteger, then Fraction, then Float. Dividing by zero signals ZeroDivide.

Large integers

Large integers are held as decimal strings and arithmetic on them is done digit by digit in CREXX. They are exact but slower than SmallIntegers. Results that fit in a SmallInteger are always converted back.

```
30 factorial printNl.
(100 factorial / 98 factorial) printNl.
(2 raisedTo: 100) printNl.
(12345678901234567890 * 98765432109876543210) printNl.

265252859812191058636308480000000
9900
1267650600228229401496703205376
1219326311370217952237463801111263526900
```

Figure 19. Factorial

Floating-point numbers

Floats print with up to 15 significant digits and always show a decimal point. The functions sqrt, sin, cos, tan, arcTan, arcSin, arcCos, ln, log, log: and exp are available; raisedTo: with a fractional exponent gives a Float. Float pi and Float e give the constants.

Reference

Selector	Meaning
+ - * / // \\ quo: rem:	Arithmetic
< > <= >= = ~=	Comparison; = between a number and a non-number answers false
abs negated sign squared reciprocal	Unary arithmetic
even odd isZero isPrime isInteger isFloat isFraction isNumber	Tests
max: min: between:and:	Comparison helpers
truncated floor ceiling rounded roundTo: fractionPart integerPart	Conversion to integers
asFloat asInteger asString asCharacter printString printString: base radix: printPaddedWith:to:	Conversion and printing
sqrt sin cos tan arcTan arcSin arcCos ln log log: exp raisedTo: **	Mathematical functions
gcd: lcm: factorial	Integer functions
bitAnd: bitOr: bitXor: bitShift: << >> bitAt:	Bit operations on non-negative integers
to: to:by: to:do: to:by:do: timesRepeat:	Intervals and loops
@	Make a Point

Figure 20. Methods of Number

Chapter 6. Collections

The collection classes

All collections answer the messages of Collection (the figure below). The classes differ in how elements are stored and retrieved.

Class	Description
Array	Fixed size, indexed from 1. Literal: #(1 2 3). Created with Array new: n, with:with:, withAll:.
OrderedCollection	Grows at both ends: add:, addFirst:, removeFirst, removeLast, add:beforeIndex:, removeIndex:.
SortedCollection	An OrderedCollection that keeps its elements in the order of a sort block (default <=).
Interval	The numbers from a start to a stop with a step: 1 to: 10, 10 to: 1 by: -2. Elements are computed, not stored.
Set	Unordered collection without duplicates.
Bag	Collection that counts the number of occurrences of each element.
Dictionary	Maps keys to values. Association is a key-value pair.
String, Symbol	Collections of characters (Chapter 7).

Figure 21. The collection classes

Protocol common to all collections

Selector	Meaning
size isEmpty notEmpty includes: occurrencesOf:	Size and membership
do: do:separatedBy: reverseDo: doWithIndex: keysAndValuesDo: with:do:	Enumeration
collect: select: reject: detect: detect:ifNone: inject:into: fold:	Transforming and searching
anySatisfy: allSatisfy: count: detectMax: detectMin: max min sum	Testing and totals
asArray asOrderedCollection asBag asSet asSortedCollection asSortedCollection:	Conversion
add: addAll: remove: remove:ifAbsent: removeAll:	Changing (not for Array)
first last first: last: allButFirst allButLast at:ifAbsent: indexOf:	Position (sequenceable collections)
copyFrom:to: copyWith: copyWithout: , reverse copyReplaceAll:with:	Copying (sequenceable collections)
sort sort: sorted sorted: asSortedCollection:	Sorting

Figure 22. Methods of Collection

```
| a |
a := #(5 3 9 1).
(a collect: [:x | x * x]) printNl.
(a select: [:x | x > 2]) printNl.
(a inject: 0 into: [:s :x | s + x]) printNl.
(a detect: [:x | x > 100] ifNone: ['none']) printNl.
a asSortedCollection printNl.
(a asSortedCollection: [:x :y | x >= y]) asArray printNl.

(25 9 81 1 )
(5 3 9 )
18
'none'
SortedCollection (1 3 5 9 )
(9 5 3 1 )
```

Figure 23. Enumerating

collect: on an Array gives an Array, on an OrderedCollection an OrderedCollection, on a Set a Set, on a Dictionary a Bag, and on an Interval an Array. select: and reject: give a collection of the same kind as the receiver.

Arrays and OrderedCollections

Indexes start at 1. An index outside the collection is an error of class IndexOutOfRange (Chapter 9). OrderedCollection grows as needed and removes from either end in constant time; inserting or removing in the middle moves the elements after it.

```
| q |
q := OrderedCollection new.
q add: 1; add: 2; add: 3.
q addFirst: 0.
q removeFirst printNl.
q removeLast printNl.
q printNl.
q removeFirst; removeFirst.
q removeFirst.

0
3
OrderedCollection (1 2 )
Object: OrderedCollection () error: access beyond end of object
OrderedCollection>>removeFirst
UndefinedObject>>executeStatements
```

Figure 24. An OrderedCollection used as a queue

Dictionaries

Keys may be numbers, strings, symbols, characters, arrays of those, or any object (compared by identity). Dictionary keys, values and associations are enumerated in the order in which the keys were first added.

Selector	Meaning
at: at:put: at:ifAbsent: at:ifAbsentPut: at:ifPresent:	Access; at: of a missing key is an error
add: addAll:	Add an Association (or the contents of another dictionary)
removeKey: removeKey:ifAbsent:	Remove
includesKey: includes: keyAtValue: keyAtValue:ifAbsent:	Search
keys (a Set) values (an Array) associations	Parts
do: keysDo: keysAndValuesDo: associationsDo:	Enumeration; do: iterates over values
collect: (a Bag) select: reject: keysSelect:	Transformation; select: answers a Dictionary
size isEmpty copy	Size and copying

Figure 25. Methods of Dictionary

```
| d |
d := Dictionary new.
d at: 'one' put: 1; at: 'two' put: 2; at: 'three' put: 3.
d printNl.
(d at: 'two') printNl.
(d at: 'four' ifAbsent: [0]) printNl.
d removeKey: 'two'.
(d collect: [:v | v * 10]) printNl.
d keys asSortedCollection printNl.
d at: #missing.

a Dictionary ('one'->1 'two'->2 'three'->3 )
2
0
Bag (10 30 )
SortedCollection ('one' 'three' )
Object: #missing error: key not found
Dictionary>>at:
UndefinedObject>>executeStatements
```

Figure 26. A dictionary

Sets and Bags

```
| s b |
s := Set new.
s add: 1; add: 2; add: 1; add: 'a'; add: 'a'.
s size printNl.
s remove: 1.
(s includes: 1) printNl.
b := Bag new.
b add: 'x'; add: 'y'; add: 'x'.
(b occurrencesOf: 'x') printNl.
b size printNl.

3
false
2
3
```

Figure 27. Set and Bag

Intervals

1 to: 5 and 10 to: 1 by: -3 create Intervals. An Interval is a collection: it understands do:, collect:, select:, inject:into:, reverse, asArray and the rest. It prints like an Array.

Chapter 7. Strings, Symbols and Characters

Strings

A String is an indexed collection of Characters. String literals are in single quotes, with an embedded quote written twice. Strings are mutable: at:put: changes a character in place. copy makes a separate string. The comparison messages <, <=, > and >= compare character codes; = compares the characters and is false between a String and a Symbol with the same letters.

Selector	Meaning
size isEmpty notEmpty first last at: at:put:	Basic access
, copyFrom:to: copyFrom: first: last: copyWith:	Joining and slicing
asUppercase asLowercase reverse reversed trimSeparators	Transforming
indexOf: occurrencesOf: includes: includesSubstring: startsWith: endsWith: indexOfSubCollection:	Searching
copyReplaceAll:with: replaceAll:with: copyReplacing:with: bindWith: %	Replacing
subStrings subStrings: lines tokenize:	Splitting; the result is an Array of Strings
asSymbol asString asNumber asInteger asArray asBag asSet asOrderedCollection	Conversion; asNumber and asInteger answer nil for text that is not a number
do: collect: select: detect: inject:into: reverseDo: keysAndValuesDo:	Enumeration; collect: must answer Characters
displayString printString displayNl printNl	Printing: displayString has no quotes
join:	The receiver is put between the elements of the argument

Figure 28. Methods of String

```
| s |
s := 'Hello, World'.
s size printNl.
s asUppercase printNl.
s reverse printNl.
(s copyFrom: 1 to: 5) printNl.
(s indexOf: $o) printNl.
(s subStrings: ', ') printNl.
(s copyReplaceAll: 'World' with: 'There') printNl.
('hello' select: [:c | c isVowel]) printNl.
'it''s' printNl.
'it''s' displayNl.

12
'HELLO, WORLD'
'dlroW ,olleH'
'Hello'
5
('Hello' 'World' )
'Hello, There'
'eo'
'it''s'
it's
```

Figure 29. Strings

Symbols

A Symbol is a String that is unique: two symbols with the same characters are the same object, so symbols compare with == and are cheap to use as keys. #foo, #at:put: and #+ are literals; asSymbol makes a symbol from a string. A symbol prints with a leading #, and with quotes when it has characters that a selector cannot

have. Symbols cannot be changed. A Symbol answers value: by sending itself, so #(1 2 3) collect: #squared works.

Characters

A Character is written $a. Character value: n gives the character with code n; Character nl, tab, space and cr give the usual ones. A Character answers value (the code), asUppercase, asLowercase, isLetter, isDigit, isVowel, isUppercase, isLowercase, isSeparator, digitValue and asString, and compares with <, >, <= and >=. Characters are unique objects.

```
$a printNl.
$a value printNl.
97 asCharacter printNl.
$a asUppercase printNl.
$5 digitValue printNl.
Character nl printNl.
Character space printNl.

$a
97
$a
$A
5
Character value: 10
$
```

Figure 30. Characters

Chapter 8. Streams and the Transcript

The Transcript

Transcript is the standard output. It understands show:, showCr:, display:, print:, displayNl:, nl, cr, tab, space, tab: and space:, and nextPutAll: and nextPut: like any stream. show: takes a String; display: and print: take any object. Output printed with printNl and displayNl goes to the same place, in order.

```
Transcript show: 'Total: '; show: 42 printString; cr.
Transcript display: 'text'; tab; print: 'text'; cr.
Transcript showCr: 'done'.

Total: 42
text  'text'
done
```

Figure 31. Writing to the Transcript

Read streams

ReadStream on: aCollection reads the elements of an Array, String or other sequenceable collection in order.

Selector	Meaning
next next: peek atEnd	Read one element, several elements, look at the next element, test for the end. next at the end is an error.
upTo: upToEnd	Read up to (and past) an element, or to the end
skip: skipSeparators position position: reset setToEnd	Move
do: collect: select:	Enumerate the rest of the stream
contents size isEmpty	The whole collection

Figure 32. Methods of ReadStream

Write streams

WriteStream on: String new collects characters; WriteStream on: Array new collects any objects. contents answers what has been written. A write stream accepts nextPut:, nextPutAll:, print:, display:, <<, nl, tab and space. It is how printOn: methods build their text.

```
| ws rs |
ws := WriteStream on: String new.
#('alpha' 'beta' 'gamma') do: [:w | ws nextPutAll: w] separatedBy: [ws nextPutAll: ', '].
ws contents displayNl.
ws := WriteStream on: Array new.
1 to: 5 do: [:i | ws nextPut: i * i].
ws contents printNl.
rs := ReadStream on: 'one two three'.
[rs atEnd] whileFalse: [(rs upTo: $ ) displayNl].

alpha, beta, gamma
(1 4 9 16 25 )
one
two
three
```

Figure 33. Streams

Standard input

Smalltalk readLine answers the next line of standard input as a String, or nil at the end of the input; Smalltalk stdinAtEnd tests for the end. The example wordfreq.st reads text this way.

Chapter 9. Exceptions

Handling

aBlock on: ExceptionClass do: [:e | ...] evaluates the block. If an exception of that class (or a subclass) is signaled while it runs, the handler block is evaluated with the exception as its argument. The value of the handler block becomes the value of the on:do: expression. A handler for several classes uses a set: on: ZeroDivide, MessageNotUnderstood do: [...].

```
| r |
r := [10 / 0] on: ZeroDivide do: [:e | e messageText displayNl. -1].
r printNl.
r := [nil foo] on: MessageNotUnderstood do: [:e | e return: 3].
r printNl.
r := [#(1 2) at: 5] on: Error do: [:e | e messageText].
r displayNl.

The program attempted to divide a number by zero
-1
3
Index value 5 out of range: only 2 elements
```

Figure 34. Catching an error

Messages understood by an exception

Selector	Meaning
messageText description	The text given to signal:, or the description of the class
return return:	Leave on:do: with nil or a value
retry retryUsing:	Run the protected block again; or leave with the value of another block
resume resume:	Continue after the signal with a value. Only resumable exceptions (not Error) may be resumed.
pass outer signal	Pass to the enclosing handler (pass does not come back; outer does if the outer handler resumes); signal again
isResumable receiver class	Query

Figure 35. Methods of Exception

Signaling

Error signal: 'text', Error new signal: 'text' and self error: 'text' signal an Error. ExceptionClass signal and signal: are class-side shortcuts. The exception classes are the following.

```
Exception
  Error
    ArithmeticError
      ZeroDivide
    MessageNotUnderstood
    InvalidValue
      InvalidArgument
        IndexOutOfRange
        NotFound
      WrongClass
    EmptyCollection
  Warning
```

Figure 36. The exception classes

An Error is not resumable. A Warning and a plain Exception are resumable: if no handler is active, a Warning prints its trace and continues, returning nil from signal.

Cleaning up

aBlock ensure: cleanup evaluates cleanup after aBlock, however aBlock ends: normally, by an exception or by a non-local return. aBlock ifCurtailed: cleanup evaluates cleanup only if aBlock does not finish normally.

```
| n r |
r := [(Warning signal: 'careful') printNl. 'done'] on: Warning do: [:e | e resume: 42].
r printNl.
n := 0.
r := [n := n + 1. n < 3 ifTrue: [Error signal: 'again']. n] on: Error do: [:e | e retry].
r printNl.
r := [[Error signal: 'inner'] ensure: ['cleanup' displayNl]] on: Error do: [:e | 'handler'
displayNl. 9].
r printNl.

42
'done'
3
handler
cleanup
9
```

Figure 37. Resume, retry and ensure

Notice that the handler runs before the stack is unwound, so the handler text appears before the cleanup text.

Your own exceptions

```
Error subclass: InsufficientFunds [
    | needed |
    needed [ ^needed ]
    needed: n [ needed := n ]
    messageText [ ^'Insufficient funds: need ', needed printString, ' more' ]
]
[InsufficientFunds new needed: 430; signal]
    on: InsufficientFunds do: [:e | e messageText displayNl].
InsufficientFunds new needed: 5; signal.

Insufficient funds: need 430 more
Object: nil error: Insufficient funds: need 5 more
UndefinedObject>>executeStatements
```

Figure 38. A user-defined exception

Chapter 10. Reflection and the System

Asking about objects

Selector	Meaning
class	The class of the receiver
isKindOf: isMemberOf: respondsTo:	Class membership; whether a message is understood
isNil notNil isString isSymbol isNumber isInteger isCharacter isArray isClass	Type tests
== = ~= ~~ hash identityHash	Identity and equality
copy shallowCopy deepCopy postCopy	Copying; copy sends postCopy to the shallow copy
perform: perform:with: perform:withArguments:	Send a message chosen at run time
instVarAt: instVarAt:put:	Access an instance variable by position
yourself ->	The receiver; make an Association

Figure 39. Reflection messages

Asking about classes

Selector	Meaning
name printString	The name of the class
superclass subclasses allSubclasses allSuperclasses	Position in the hierarchy
selectors includesSelector: canUnderstand:	The methods of a class
instanceVariableNames allInstVarNames instSize	Instance variables
inheritsFrom: includesBehavior:	Inheritance tests
subclass: subclass:instanceVariableNames:classVariableNames:package:	Create a subclass
addInstVarName: addClassVarName: instanceVariableNames:	Add variables

Figure 40. Class messages

```
Object subclass: Probe [ |a| a [ ^a ] ]
Probe selectors printNl.
Probe instanceVariableNames printNl.
(Probe new respondsTo: #a) printNl.
(Probe new respondsTo: #zork) printNl.
(3 perform: #+ with: 4) printNl.
(Probe new isKindOf: Object) printNl.
Probe superclass printNl.

(#a )
(#a )
true
false
7
true
Object
```

Figure 41. Reflection

The Smalltalk dictionary

Smalltalk is the dictionary of global variables. Smalltalk at: #Name put: value creates or sets a global, Smalltalk at: #Name reads it, and Smalltalk includesKey: #Name tests for it. A global created this way can

then be used by its plain name, Name, in any program. Smalltalk version answers the version string. Smalltalk halt and Smalltalk quit: n end the program.

Chapter 11. Errors and Traces

Unhandled errors

When an exception is signaled and no handler is active, the interpreter prints a line that names the object and the error, and then a trace of the active methods, innermost first, ending with UndefinedObject>>executeStatements. The statement is abandoned and the program continues with the next one. This is the form used by GNU Smalltalk.

```
Object subclass: T [
    a [ ^self b ]
    b [ ^1 / 0 ]
]
T new a.
'next' displayNl.

Object: 1 error: The program attempted to divide a number by zero
SmallInteger(Number)>>zeroDivide
SmallInteger(Number)>>/
T>>b
T>>a
UndefinedObject>>executeStatements
next
```

Figure 42. A trace

A trace line reads Receiver(ImplementingClass)>>selector. When the receiver's class is also the class that defines the method, the part in parentheses is left out. Blocks that were compiled in line, and the lines of the exception machinery itself, do not appear. ensure: blocks run after the trace is printed.

Common messages

Message	Cause
did not understand #selector	A message was sent that the receiver does not understand. For an argument of an arithmetic message the missing selector is #generality.
The program attempted to divide a number by zero	/ // \\ rem: or quo: with a zero divisor
Index value n out of range: only m elements	at: or at:put: with an index outside a collection
key not found	Dictionary at: or removeKey: of a missing key
access beyond end of object	removeFirst, removeLast or first on an empty OrderedCollection
This method is a subclass responsibility	subclassResponsibility was sent
wrong argument count	A block was evaluated with the wrong number of arguments
call stack too deep	More than 20000 nested method calls
BlockContext>>cannotReturn:	A block with ^ was evaluated after its method had returned

Figure 43. Error messages

Parse errors

A syntax error is reported as Parse error (file line n): text. The interpreter then skips to the end of the statement and goes on.

Chapter 12. How Smalltalk/CREXX Works

Objects

A reference to an object (an oop) is an integer. An odd integer 2n+1 stands for the SmallInteger n and has no storage. An even integer 2i is the index i of an entry in the object table, which is a set of parallel arrays: class, kind, number of slots, start of the slots in a common slot pool, a numeric field, a string field and a float field. nil, true and false are entries 1, 2 and 3.

The kind of an entry says how the native fields are used: plain object, String, Symbol, Float, Array, Character, block, large integer, Dictionary, class or metaclass. A String holds its text in the string field, so that string operations are done by CREXX built-in functions, and a large integer holds its decimal digits in the same field. A Dictionary holds a hash map from key text to a linked entry, which gives constant-time access and keeps insertion order.

Classes and methods

Each class has a number. Its record holds the superclass, the metaclass, the names of the instance variables including those it inherits, and the kind of instances it creates. Methods are kept in a table indexed by class number and selector number. Each selector name gets a number the first time it is seen. A cache of lookup results is flushed whenever a method is installed.

The resolver

A method body is parsed into a raw tree. The resolver walks it once and replaces names by their meaning: a local variable becomes (levels up, slot), an instance variable becomes its position in the object, a class variable or global becomes a cell number. The control-flow messages with literal block arguments become single nodes, and their variables live in the frame of the enclosing method or block, so that no block object is created for them.

The evaluator

Frames are kept in one array. A frame holds the outer frame, the home (method) frame, self, the method, and then the arguments and temporaries. Frames of methods and blocks that create no closures are returned to a free list as soon as they finish; the others are left in place because a block may still refer to them. The arguments of a send are placed on a separate stack.

Returns and exceptions are done by unwinding. The evaluator checks a global flag after every call; a ^ in a block sets it to return-to-this-frame and the frames above are abandoned in turn. Exceptions use the same mechanism: signal finds a handler by searching a stack of active handlers and runs its block on top of the stack that signaled, so that the handler can resume; return, retry and pass then unwind to the on:do: that installed the handler.

Primitives

A method may start with <primitive: name>. The named CREXX routine is run first; if it returns a value that is the result of the method, and if it fails the Smalltalk statements that follow the pragma are run. This is how the library provides, for example, SmallInteger + with fallbacks for fractions and errors. Arithmetic and comparison of two SmallIntegers is done directly by the send routine without looking up a method.

Adding a primitive

1. Add the name to src/prims.txt. Its position gives it a number, and the constant PR_NAME.
2. Add a branch for PR_NAME to prim_a, prim_b or prim_c in src/f_prim.crexx. The receiver is rcv, the arguments are ast[base+1] to ast[base+na]. Return 0 to fail.
3. Use it in the library: method [<primitive: name> ^fallback].
4. Run python3 build.py and tests/run_tests.sh.

The build

build.py joins the files in src in a fixed order, replaces the symbolic constants (token types TK_, node types ND_, object kinds KD_, unwind kinds UW_, selectors S_, primitives PR_) by numbers, fills in the list of global variables of the namespace, and writes smalltalk.crexx.

Chapter 13. Restrictions and Known Problems

- **new does not call initialize.** Override the class-side new, as the examples do.
- **Dictionary keys** are compared by built-in equality for numbers, strings, symbols, characters and arrays of those, and by identity for other objects. A user-defined = and hash are not consulted by Dictionary, Set or Bag.
- **No garbage collector.** Objects are never freed. A program that creates many millions of objects will use a great deal of memory.
- **Speed.** The interpreter is a program running on a virtual machine. About 300 thousand simple sends per second can be expected; a million-iteration loop takes a few seconds. Large-integer arithmetic works on decimal strings and is slow for numbers of thousands of digits.
- **Floats** have the precision of the CREXX floating-point type, about 15 digits, and print with that many. The printed form of results such as 0.1 + 0.2 is 0.3.
- **SmallInteger** covers 18 decimal digits (SmallInteger maxVal is 999999999999999999); larger values are large integers.
- **No processes** (fork, Semaphore, Process), no Date and Time classes beyond Time millisecondClock and millisecondsToRun:, no file streams, no thisContext.
- **Class definitions** must name an existing superclass. The bang chunk format of old Smalltalk-80 files is not supported.
- **Namespaces.** A dotted name such as SystemExceptions.NotFound is read as the last part only.
- **Strings** are byte strings: characters have codes 0 to 255.
- **String comparison** with < is by character code, not case-insensitive.

Appendix A. Command Summary

```
./smalltalk [file.st ...]
crexx smalltalk.crexx -nocompile -args -L lib [file.st ...]

-L dir   directory of the library files (the launcher supplies it)
-h       show usage
```

Figure 44. Command line

With no file name the program is read from standard input. The environment variable CREXX names the crexx driver used by the launcher.

Appendix B. Contents of the Repository

Path	Contents
smalltalk	Launcher script.
smalltalk.crexx	The interpreter, generated by build.py (checked in so that the repository runs at once).
build.py	Assembles smalltalk.crexx from src.
src/	The interpreter sources, consts.txt, selectors.txt and prims.txt.
lib/	The class library in Smalltalk: kernel.st, numbers.st, collections.st, streams.st, exceptions.st.
examples/	Sample programs.
tests/	Regression tests and run_tests.sh.
docs/	This manual as a Word document and as a PDF.
README.md	Short description of the project.

Figure 45. Repository

Appendix C. Class Hierarchy

```
Object
  Behavior
    ClassDescription
      Class
      Metaclass
  Boolean
    True
    False
  UndefinedObject
  Magnitude
    Character
    Number
      Integer
        SmallInteger
        LargeInteger
          LargePositiveInteger
          LargeNegativeInteger
      Float
        FloatD
      Fraction
    LookupKey
      Association
    Date, Time
  Collection
    SequenceableCollection
      ArrayedCollection
        Array
        CharacterArray
          String
            Symbol
        Interval
      OrderedCollection
        SortedCollection
    HashedCollection
      Dictionary
        IdentityDictionary
      Set
        IdentitySet
      Bag
  Stream
    PositionableStream
      ReadStream
      WriteStream
        ReadWriteStream
    TextCollector
  BlockClosure, Message, Point, Random, SystemDictionary
  Signal
    Exception
      Error, Warning
```

Figure 46. The kernel classes

Exception classes are shown in Chapter 9.

Appendix D. Sample Programs

The directory examples holds the programs in the figure below.

File	Shows
hello.st	Transcript, printNl
fib.st	Class-side methods, recursion, large integers
shapes.st	Inheritance, polymorphism, printOn:, sorted collections, fractions
bank.st	User-defined exceptions, on:do:, ensure:, error:
primes.st	Array, loops, select: on an Interval
hanoi.st	Recursive blocks
streams.st	WriteStream, ReadStream, strings
wordfreq.st	Dictionary, standard input, sorting with a sort block

Figure 47. Sample programs

The regression tests tests/01_classes_basic.st to tests/07_objects.st exercise classes, collections, strings, numbers, exceptions and the object model, and show the exact output, traces included.